

FitBuddy

AI Fitness Plan Generator using Gemini Models

Personalized • Intelligent • Easy to Follow

TEAM MEMBERS

Team Leader: Kabila S

Jothikrishnan K

Kalaiyarasan E

Kamaraj K

Kanishka M



What is FitBuddy?

FitBuddy is an AI-powered fitness plan generator. Users provide basic information such as goal, fitness level, available time and preferences. Gemini processes the information and generates a structured fitness plan.



Problem

Generic workout plans may not match individual goals, schedules or experience.



Solution

FitBuddy creates personalized recommendations dynamically and presents them in an easy-to-follow format.

2. Objectives

● Personalization

Generate plans based on individual inputs.

● Automation

Reduce manual effort in creating routines.

● AI Assistance

Use Gemini to interpret goals and preferences.

● Usability

Provide a simple and readable plan.

● Adaptability

Allow plans to be regenerated as goals change.

● Progress Focus

Encourage consistent tracking and improvement.

3. System Workflow

From user profile to an AI-generated fitness plan.



◆ Core Idea

The system converts structured user information into a clear plan through an AI prompt and Gemini response.

4. User Inputs

Example information collected before generating a plan.



Goal

Weight management / strength / general fitness



Fitness Level

Beginner / intermediate / advanced



Schedule

Available days and session duration



Preferences

Home or gym, equipment availability



Lifestyle

Activity level and basic preferences



Prompt Engineering

User inputs are transformed into a structured prompt with goals, constraints and desired output format.



Gemini Model

Gemini analyzes the prompt and generates a tailored response containing workout and guidance sections.



Response Handling

The application formats the AI response into a clean plan that users can read and follow.

6. Sample Generated Plan

Illustrative output structure — actual recommendations should be personalized.

●	MON	●	TUE	●	WED	●	THU	●	FRI	●	SAT	●	SUN
	Full Body		Cardio		Recovery		Upper Body		Lower Body		Light Activity		Rest

✓ Plan Sections

Warm-up • Main workout • Cool-down • Recovery notes • General nutrition guidance

7. Technology Stack



Frontend

HTML • CSS • JavaScript



Backend

Java / Spring Boot or Python backend



AI Layer

Google Gemini Models



Database

MySQL / optional user profile storage



API

Gemini API



Tools

STS / VS Code • GitHub



Advantages

- Personalized output
- Fast plan generation
- Easy to update
- Conversational AI experience
- Can be extended with progress tracking



Important Limitations

- AI output should be reviewed
- Not a substitute for medical advice
- User data privacy must be protected
- Exercise plans should consider safety and individual conditions

→ **Wearables**

Integrate activity and heart-rate data.

→ **Progress Tracking**

Track completed workouts and milestones.

→ **Voice Assistant**

Allow voice-based fitness interactions.

→ **Adaptive Plans**

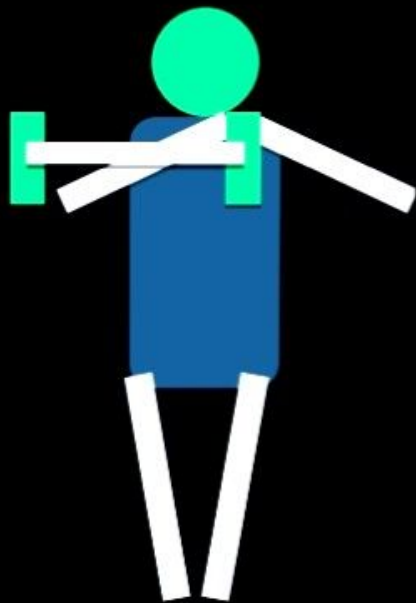
Automatically adjust plans using progress data.

→ **Nutrition Module**

Add meal-planning and habit guidance.

→ **Mobile App**

Extend FitBuddy to Android/iOS.



★ Conclusion

FitBuddy demonstrates how Gemini-based generative AI can be used to create personalized fitness plans from structured user inputs. The system combines a simple user experience with AI-generated guidance and provides a strong foundation for future features such as progress tracking, wearables and mobile applications.

THANK YOU

FitBuddy - AI Fitness Plan Generator using Gemini Models

